Noise Removal versus Signal Loss in Network Filtering of Daily GNSS Time Series: Internal Working Report

H. Oğuz Bolat

[a] *Department of Geomatics Engineering, Yıldız Technical University, Istanbul, Türkiye*

Abstract

This report describes, in full, a signal-loss audit of network filters for daily GNSS position time series, reports its results and discusses them. Daily IGS20 solutions of the Nevada Geodetic Laboratory for 1,251 stations in and around California are reduced to residuals by removing a trajectory model fitted on 2008–2019. Seventeen filter configurations from three families are audited: target-inclusive filters, which use the target station in its own estimate (network stack, sub-regional probabilistic PCA, robust weighted median); distance-excluded filters, which use every station beyond a radius R; and leave-one-out filters over nearest neighbours, from weighted means and ridge regression to a learned kernel and a space–time attention network, the last also trained with planted transients. Each filter is scored on two noise measures, the reduction in short-period scatter and the leave-one-out reconstruction error, and on two signal measures, the fraction of a planted transient that it keeps and the spurious signal it creates outside the transient. Transients are planted as idealised Gaussian fields of 10–100 km width and as fault-slip events on the SCEC Community Fault Model; real earthquake offsets and the Ridgecrest postseismic displacement are scored as well. On the test years (2022 to September 2026), the mean of every station beyond R (far stack) removes 22.7% of the short-period scatter at $R = 0$ and keeps 0.98 and 0.83 of 25- and 100-km transients; the weighted mean of the 16 nearest stations removes 14.1% and keeps 0.25 and 0.00. The reconstruction errors of the attention network, its transient-trained version and the learned kernel over 256 neighbours are not distinguishable from that of the far stack, and each keeps less planted signal. Results on the validation years (2020–2021)

Email address: oguz.bolat@std.yildiz.edu.tr (H. Oğuz Bolat)

are given alongside where they differ. The far stack's loss of planted signal is explained by the share of a transient that enters the network mean; nearest-neighbour filters keep fault-slip signal near the fault and remove it farther away; and scatter reduction, velocity repeatability and reconstruction error all fail to reveal these losses, so signal kept and spurious signal need to be reported with them.

Keywords: GNSS time series, common-mode error, spatial filtering, transient deformation, signal preservation, deep learning

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Status of this document. This is an internal working report, written as a reference while the pilot paper (`paper/pilot.tex`) is being written. Its text was drafted by Claude Opus 5.5 (Anthropic) through Claude Code, at the author's request, from the repository, its design documents and its results files; every number was taken from those files on 14 October 2026. Sections 1 to 6 (literature, data, method, filters, results) state what was done and what was measured. The Discussion and Conclusions (Sections 7 and 8), added on 16 October 2026, are the tool's own interpretation, written so that the study can be read as a whole; they are not the author's interpretation. It is not a manuscript, and its wording is not intended to be reused in one. A post-freeze deep-learning experiment started on 14 October 2026 is not described.

1. Introduction

GNSS stations record crustal deformation together with noise. Part of the noise is common to many stations of a network (common-mode error), and arises from errors in satellite orbits and clocks, in the realisation of the reference frame and from large-scale environmental loading (Wdowinski et al., 1997; Dong et al., 2006; Kreemer and Blewitt, 2021). Common-mode error has been estimated by stacking (Wdowinski et al., 1997), by principal or independent component analysis (Dong et al., 2006; Gruszczynski et al., 2018; Zhou et al., 2022), and by weighted means and medians of neighbouring stations (Tian and Shen, 2016; Kreemer and Blewitt, 2021). The evaluation reported for these filters is most often the reduction in the root-mean-square (RMS) scatter of the filtered series (Tian and Shen, 2016; Kreemer and Blewitt, 2021; Zhou et al., 2022). Kreemer and Blewitt (2021) note that the main reason their filter reduces scatter more than traditional stacking is that it absorbs local common-mode transients.

Learned filters have been evaluated in other ways. Bachelot et al. (2025) predict each station from its eight nearest stations beyond 400 km, a distance chosen empirically in their study area as the shortest at which slow-slip offsets were no longer reduced by denoising, and they assess signal preservation on 280 slow-slip offsets measured on raw and denoised series. Costantino et al. (2024) add synthetic slow-slip events to noise generated from real GNSS series and report the error of the recovered displacement against the signal-to-noise ratio. Mastella et al. (2025) train a single-station method on synthetic series and evaluate the noise removed from real series against the residual of a trajectory-model fit (Bedford and Bevis, 2018). Tian and Shen (2016) estimated the common mode from stations beyond increasing distance thresholds and used the threshold at which the RMS changed most sharply as the approximate extent of a regional transient. Gualandi et al. (2016) evaluated source separation on simulated networks with known sources in synthetic noise. In Cascadia, Costantino et al. (2026) report that the scaling, b-value and stress-drop distribution of slow-slip events change with the detection threshold applied to slip inverted from denoised data, and the debate over a reported precursory acceleration of slip (Bletery and Nocquet, 2023, 2025) turned on whether a common-mode correction removes only noise.

The study described here measures signal loss directly. Transients of known extent, amplitude and duration are planted in the real residuals of 1,251 stations; each filter is run on the residuals with and without them; and the difference between the two cleaned series gives the fraction of each transient that survives. The same runs give the noise each filter removes and the spurious signal it creates where nothing was planted. Each filter predicts a station from N other stations, excluding every station within a radius R of it; both N (from 16 stations to the whole network) and R (from 0 to 600 km) are treated as experimental variables. Section 2 reviews the literature, Section 3 describes the data and their processing, Section 4 the audit, Section 5 the filters (Table 2), Section 6 the results, Section 7 discusses them and Section 8 concludes.

2. Literature review

This section is condensed from the research proposal. Two works cited there are left out: a blog post, and a study the author has not read in full.

2.1. Classical common-mode filtering

Network filtering of GNSS data began with the Southern California Integrated GPS Network, where the common-mode error was estimated as the weighted mean of the residuals at the other stations and subtracted from each series (Wdowinski et al., 1997). Principal component analysis and the Karhunen–Loève expansion extended this to common modes that vary in space (Dong et al., 2006). Later work made principal component filtering applicable to networks whose stations start and stop at different times: a modified analysis that reproduces each series from its principal components without filling gaps (Shen et al., 2014), probabilistic principal component analysis that treats missing days as latent values (Gruszczynski et al., 2018), and a variational Bayesian version that also selects the number of components (Li et al., 2020). Because the common mode varies in space, large networks have been divided into sub-regions; on 208 Chinese stations, adaptive sub-regional filtering reduced residual scatter by 20–29%, against 15–18% for a single global analysis (Liu et al., 2026). Independent component analysis offers an alternative decomposition (Gualandi et al., 2016; Zhou et al., 2022).

Tian and Shen (2016) weighted each neighbouring station by its distance and by the correlation of its residuals with the target, chose the weights by a grid search that minimises the residual scatter after filtering, and outperformed principal component analysis by more than 13% in scatter reduction on about 1,600 stations in western North America. They also stepped through minimum-distance thresholds to separate long-range from regional common modes and to estimate the extent of regional transients. The common-mode component imaging of Kreemer and Blewitt (2021) takes a correlation-weighted median over nearby stations, including the target itself when it qualifies, and reduced residual scatter by 39–50% on about 2,500 stations in western Europe. Across this family, performance is reported as scatter reduction, reduced inter-station correlation, velocity uncertainty or the recovery of a simulated common mode.

2.2. Distance-excluded and learned filters

The graph neural network of Bachelot et al. (2025) represents stations as nodes and connects each to its eight nearest stations beyond 400 km. During training, 30% of the nodes are masked and the error is computed on the masked nodes only, so the network learns to predict a station from distant stations, and the prediction is subtracted as noise. They compared it with three stacks, including a stack of all stations beyond 400 km, which performed

comparably and better on the network-average metric. The 400 km was chosen as the shortest distance at which 280 slow-slip offsets from 15 Cascadia events were no longer reduced by denoising.

Costantino et al. (2024) train a network of graph recurrent layers and transformers to extract slow slip from synthetic examples, slip on the subduction interface added to noise synthesised from real Cascadia series by principal component decomposition and Fourier phase randomisation; validated by synthetic recovery and agreement with tremor, it was later used to invert fifteen years of denoised data (Costantino et al., 2026). Mastella et al. (2025) model high-frequency scatter at single stations, with noise defined as the residual of a trajectory decomposition (Bedford and Bevis, 2018) and synthetic training noise from an adversarial network. Teutschmann et al. (2026) pretrained a self-supervised foundation model on 17,651 NGL stations by masked latent prediction, station by station, and fine-tuned it for displacement forecasting and the localisation of seismic offsets.

Predicting a withheld sensor from its neighbours is established elsewhere: tilt and compliance noise is removed from ocean-bottom seismometers by predicting the vertical channel from other channels (Crawford and Webb, 2000; Janiszewski et al., 2019), distributed acoustic sensing records are denoised by predicting each channel from its neighbours (Batson and Royer, 2019; van den Ende et al., 2023), and graph networks for spatiotemporal kriging are trained by reconstructing withheld sensors (Wu et al., 2021; Cini et al., 2022; Marisca et al., 2022). Excluding nearby observations so that spatial autocorrelation does not inflate apparent skill is the basis of buffered spatial cross-validation (Le Rest et al., 2014).

2.3. How signal preservation has been assessed

Three approaches recur (Table 1). Observed transients can be compared before and after filtering (Bachelot et al., 2025); this uses real signals but has no ground truth. Synthetic signals can be added to synthetic noise (Costantino et al., 2024; Mastella et al., 2025); this has ground truth but depends on how realistic the noise is, and has been analysed against signal-to-noise ratio rather than spatial scale. Independent data such as tremor can confirm that an extracted signal is real (Costantino et al., 2024), but not measure what was lost. Detectors of transients are commonly trained on synthetic signals: Costantino et al. (2023) injected 60,000 synthetic slow-slip events computed with elastic dislocations (Okada, 1992) into surrogate noise and recovered 35 of 40 catalogued events (Michel et al., 2019); other

Table 1: How signal preservation has been assessed in common-mode filtering and GNSS denoising.

Study	Test of signal preservation	Ground truth	Dependence on spatial scale
Tian and Shen (2016)	Minimum residual scatter; scatter against distance threshold	None	Transient extent inferred, loss not measured
Kreemer and Blewitt (2021)	Scatter reduction, velocity repeatability	None	Absorption of local transients noted, not measured
PCA and ICA filters[a]	Simulated common mode, inter-station correlation, velocity uncertainty	Noise only	Not examined
Mastella et al. (2025)	Synthetic trajectories in synthetic noise	Synthetic	Single station
Costantino et al. (2024)	Synthetic slow slip in synthetic noise; tremor	Synthetic	Against signal-to-noise ratio
Bachelot et al. (2025)	280 observed slow-slip offsets before and after	None	One distance (400 km)
This study	Transients planted in real noise; fault-slip events; real offsets and Ridgecrest	Planted	Retention and spurious signal against width and exclusion radius

[a] Shen et al. (2014); Gruszczynski et al. (2018); Li et al. (2020); Zhou et al. (2022); Liu et al. (2026).

approaches include training on real labelled windows (Tanaka et al., 2025), autoencoders for InSAR time series (Rouet-Leduc et al., 2021), the network strain filter (Ohtani et al., 2010), Kalman smoothing with principal components (Ji and Herring, 2013) and sparse estimation (Riel et al., 2014). The only statewide analysis of transient deformation in California is that of Klein et al. (2019).

2.4. Gaps

Four gaps motivate the study. No study has measured, with ground truth, how much of a transient a common-mode filter removes as a function of the transient's size relative to the filter's reach. The reach has been chosen

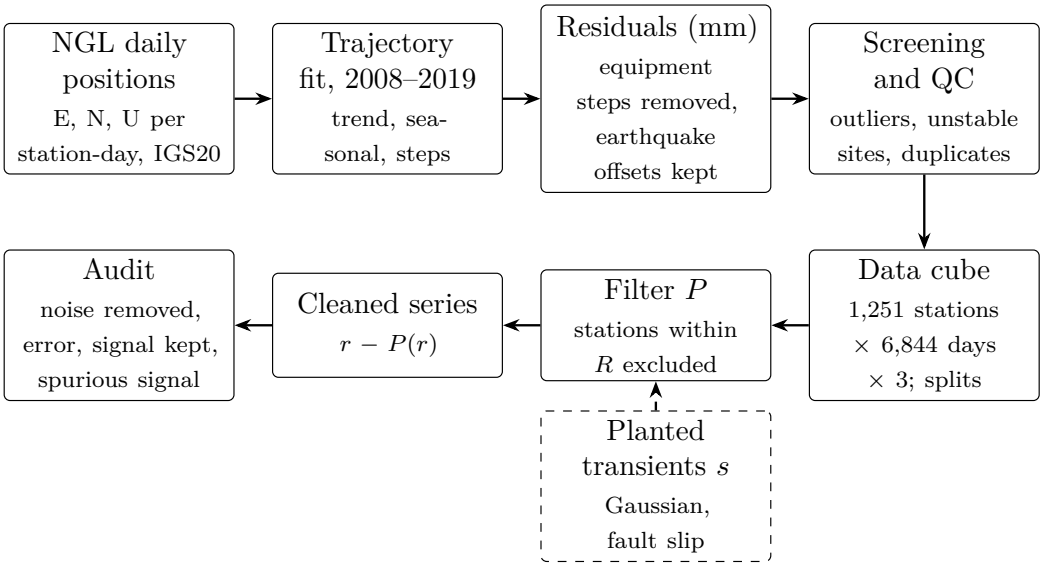


Figure 1: Processing chain from daily GNSS positions to the signal-loss audit.

empirically, as a fixed 400 km (Bachelot et al., 2025) or by scatter-based thresholds (Tian and Shen, 2016). Filters that use the target station, which include most classical methods, have not been compared with leave-one-out filters on signal loss. And learned filters have been compared with simple ones on noise reduction, not at equal signal kept.

3. Data and processing

Figure 1 shows the processing chain. Daily positions are reduced to residuals by removing each station's trajectory, screened, and stored in a single array (the cube). Every filter is applied to the cube, and the cleaned series are scored by the audit of Section 4. All processing is done by the `gnssdl` package (`gnssdl build` for the cube, `gnssdl bench` for filtering and scoring).

3.1. Data

The data are the daily IGS20 solutions of the Nevada Geodetic Laboratory (NGL) (Blewitt et al., 2018; Nevada Geodetic Laboratory, 2026). Each solution gives the east, north and up position of a station's antenna relative to a fixed reference point near the station, estimated from 24 hours of observations, with formal uncertainties of about 1 mm in the horizontal and

3–4 mm in the vertical. NGL publishes new days with a latency of about ten days. In southern California the noise in these series contains flicker and random-walk components in addition to white noise (Langbein, 2008).

All stations between 32.4° and 42.1°N and between 124.6° and 114.0°W with at least 1,095 daily solutions were taken: 1,381 stations in California and adjacent parts of Nevada, Oregon and Baja California. The analysis starts on 1 January 2008. About 390 stations reported on a typical day in 2005, and between about 820 and 1,030 have done so since 2008. Of the 1,381 stations, 130 were removed: 30 with less than one year of data after 2008; 22 with non-tectonic site motion, such as groundwater-driven subsidence in the Central Valley, identified by a robust scale of the trajectory residual (Section 3.2) above 10 mm in the horizontal or 30 mm in the vertical; and 78 co-located duplicates within 1 km of a longer-running station, of which only the station with the most data was kept. The remaining 1,251 stations (Fig. 2) give 6.45×10^6 station-days. On 1 July 2018 the median distance to the nearest station was 9.7 km, and a typical station had 22 stations within 50 km and 73 within 100 km. The 16th-nearest neighbour of a station lies at a median distance of 40 km (10th–90th percentiles 22–79 km), the 64th at 88 km and the 256th at 204 km.

3.2. Residuals

For each station i and component c, the trajectory model

$$x_{i,c}(t) = a + v\,(t - t_0) + \sum_{k=1}^{2} \left[p_k \sin(2\pi k t) + q_k \cos(2\pi k t)\right] + \sum_j b_j\, H(t - t_j) + r_{i,c}(t) \tag{1}$$

is fitted by least squares, where a is a constant, v a linear rate about the mean epoch t_0 of the fitted data, p_k and q_k the amplitudes of the annual and semi-annual harmonics, b_j the amplitudes of Heaviside steps H at the epochs t_j of the NGL step catalogue, $r_{i,c}(t)$ the residual and t the time in years (Fig. 3). Only training days (2008–2019, outside the Ridgecrest exclusion described below) enter the fit, and the fitted trajectory is extrapolated to later years. Equipment steps are removed everywhere: within the training years they are estimated in the fit, and after them by a local estimate, the median residual over the 30 days after the change minus that over the 30 days before. Earthquake steps are not removed from the residuals; those within the training years are estimated in the fit, so that they do not bias the rate, and then added back. A station with fewer than 365 training epochs is fitted

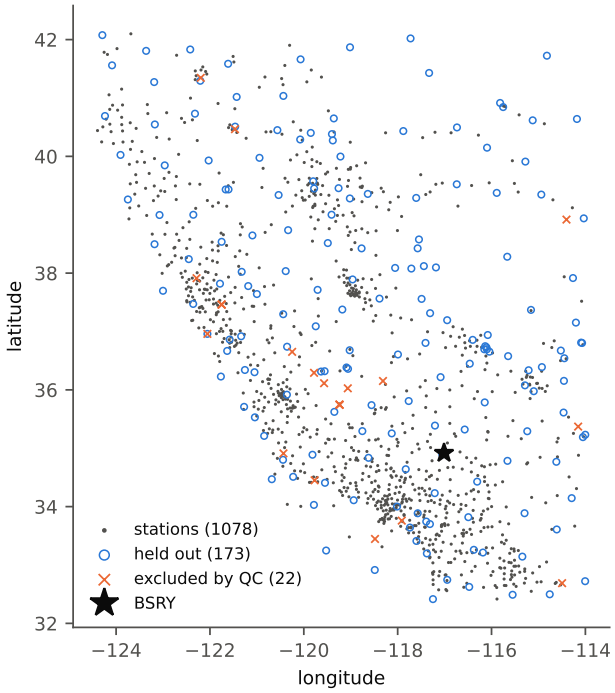


Figure 2: Stations used, held out and excluded.

on its first 730 days, or its first 365 epochs, whichever reaches further, and is held out (Section 3.3).

A value more than five robust standard deviations (1.4826 times the median absolute deviation) from its 61-day running median in any component is treated as missing; 32,544 station-days, about 0.5% of the data, are removed this way. The station QC of Section 3.1 uses a second trajectory fit from which earthquake steps are also removed, so that coseismic motion does not count against a station. Stations that develop problems after the training years are kept as inputs but not scored: those whose day-to-day scatter increases more than threefold, and those with a jump of more than 50 mm in the horizontal or 100 mm in the vertical between consecutive 30-day medians with no catalogued step within 45 days; 19 stations are excluded from scoring this way. The local equipment-step estimate and the outlier screening use days from every year, including the validation and test years, but no other station; their effect on a day that is kept is at most a constant shift or the removal of a spike.

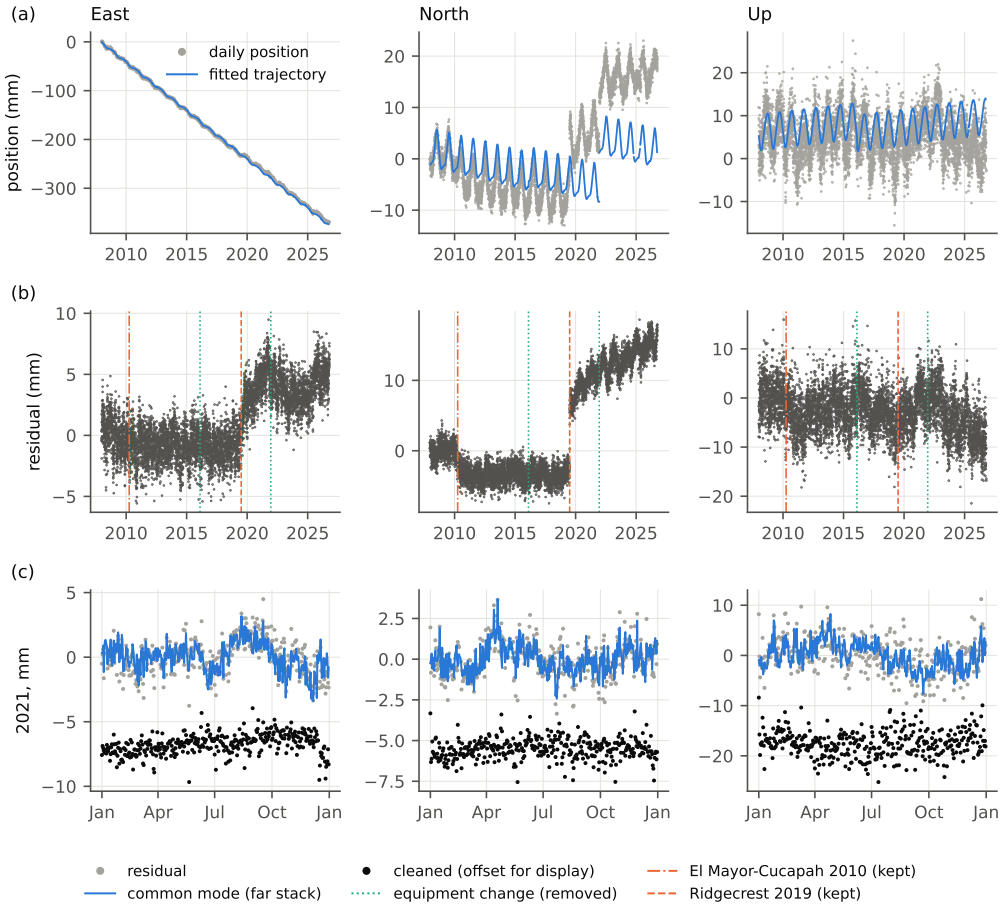


Figure 3: From positions to residuals at station BSRY: (a) positions and fitted trajectory, (b) residual, (c) one validation year with the network common mode and the cleaned series.

The residuals, their formal uncertainties and an availability mask are stored in an array of 1,251 stations by 6,844 days by three components, together with the neighbour lists of every station at each exclusion radius (the 16 nearest stations beyond R, searched to $R + 300$ km). Days are split into training (2008–2019), validation (2020–2021) and test years (2022 to 26 September 2026, the last day available when the method was frozen); days published after the freeze will form a prospective test.

3.3. Held-out stations and excluded intervals

The held-out stations are never used in training, and serve to test how well trained filters generalise to stations they have not seen. 125 stations, 10% of the network, are chosen by farthest-point sampling (Gonzalez, 1985) with seed 0, which starts from a random station and repeatedly adds the station farthest from all those already chosen. A further 56 stations are held out because they have fewer than 365 daily solutions in 2008–2019; 8 stations are in both groups, giving 173 held-out stations. The interval from 1 July 2019 to 30 June 2020 at stations within 150 km of the 2019 Ridgecrest mainshock (35.77°N, 117.60°W) is excluded from training, so that no model is fitted or trained on its postseismic signal.

3.4. Displacement rather than increments

Filters are applied to displacement rather than to daily increments. For example, 2 cm of postseismic displacement accumulated over a year is a daily increment of about 0.055 mm, whereas the day-to-day scatter of the increments is about 1.4 mm in the horizontal components. Increments are used only as an additional input to the attention network (Section 5.4).

4. The signal-loss audit

4.1. Leave-one-out filtering and the exclusion radius

A filter P estimates the part of the residual of station i that is shared with other stations, and the cleaned series is $r - P(r)$. A leave-one-out filter never reads station i itself when estimating it, and never reads any station within R km of it (Fig. 4). The target-inclusive filters of Section 5.1 read the target on purpose, as published filters of their kind do; they are applied to the same cube and scored on noise removed and on the signal measures, but not on reconstruction error. The exclusion radius R is an experimental variable: M1 is run at $R = 0$, 10, 25, 50, 100, 200 and 400 km and the far stack at 0, 25, 50, 100, 200, 400 and 600 km; the other leave-one-out filters are run at the radii in Table 4, and M2, M3 and M4 at $R = 0$ only.

Two leak checks are run for every leave-one-out filter at every radius, on the first year of the validation period. Both compare predictions on two inputs that differ only in values the filter must not read. In the radius check, the target and every station within R of it are set to a poison value of 10^6 mm, for five randomly chosen targets. In the mask check, the cells hidden for the

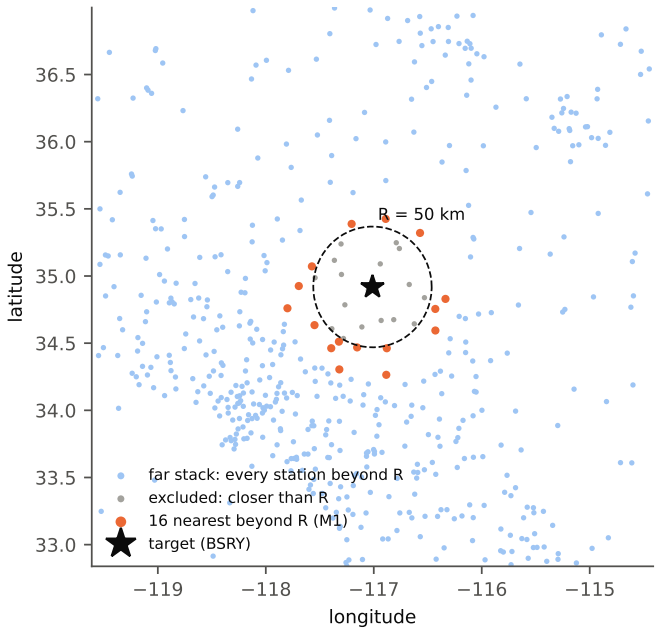


Figure 4: What a prediction for one target may use at $R = 50$ km.

mask scores below are filled once with missing values and once with the poison value, while remaining marked as unavailable. In both, the predictions must agree to within 0.001 mm. Every filter passed both checks at every radius with a largest difference of 0.0 mm. Because a filter that passes the radius check never reads the target, one prediction on the unhidden cube is a leave-one-out prediction for every station at once, and this is how all cleaned series are produced.

The harness also computes scores on fixed evaluation masks (seed 20260101): 10% of available station-days at random (`scatter`), one 30-day block per station and year (`block`), all days of the held-out stations (`station`) and the Ridgecrest window. These mask scores are not the reconstruction error of Section 4.2; they are used here only for the spread between training seeds (Section 6.7), and are labelled as such.

4.2. Measures of noise removed and of planted transients

Filters are compared on two measures of noise and two of signal, with a third, velocity repeatability, reported separately. Noise removed is the

reduction in short-period scatter at each scored station,

$$1 - \frac{\mathrm{rms}\Big(\mathrm{fast}(r - P(r))\Big)}{\mathrm{rms}\Big(\mathrm{fast}(r)\Big)}, \tag{2}$$

where the fast part of a series is the series minus its 61-day running median. It is averaged over the three components and summarised by its median over stations. Predicting zero gives 0.

Reconstruction error is the root mean square of the fast part of the leave-one-out error $r - P(r)$, normalised by each station's robust noise scale from the training years, averaged over the three components, and summarised by its median over the seen, scored stations. Days within ± 30 days of an $\mathrm{M} \geq 6$ earthquake catalogued for a station are left out. Predicting zero gives 0.753 on the test years and 0.723 on the validation years rather than 1, because separating the slow part also removes some of each station's own noise. Reconstruction error is the criterion on which the leave-one-out filters were tuned, on the validation years.

Retention measures the fraction of a planted transient that survives filtering. A transient s is added to the residuals, and both the original and the modified residuals are filtered. The change in the cleaned series and the retention are

$$\Delta = \Big(r + s - P(r + s)\Big) - \Big(r - P(r)\Big), \qquad \rho = \frac{\langle \Delta,\, s \rangle}{\langle s,\, s \rangle}, \tag{3}$$

with the inner products over the horizontal components, the days of the transient and the scored stations within $2L$ of its centre, where L is its width. Pairing with the unmodified run removes the noise term; for a linear filter, $\rho = \langle s - P(s), s \rangle / \langle s, s \rangle$ exactly. $\rho = 1$ means that the transient is kept, $\rho = 0$ that it is removed, and $\rho > 1$ that the filter adds to it. Retention is summarised by its median over transients.

Spurious signal is the change created where nothing was planted. A Gaussian transient is set to zero beyond $3L$ from its centre (its footprint). At the scored stations outside the footprint whose cleaned series the transient can change, which depend on the filter (stations with a neighbour inside the footprint for the nearest-neighbour filters, stations whose far set includes one for the far stack, every station of the same region for C3, and every station for C1), the peak horizontal amplitude of Δ is taken as a fraction of the amplitude of the transient, the largest value over these stations is kept, and

its median over transients is reported. For fault-slip events, which have no single amplitude, the stations outside are those moved by less than 0.1 mm by the event, and the reference is the event's largest displacement at a scored station.

Velocity repeatability compares the velocity of each cleaned series on 2022–2023 with that on 2024–2025. Each velocity is a MIDAS-style estimate (Blewitt et al., 2016): the median of the slopes between all pairs of days exactly 365 days apart within the window, recomputed after removing slopes more than two robust standard deviations from the first median. Pairs that span a catalogued earthquake step or touch a day left out of scoring are skipped, and at least 100 pairs are required. The measure is the median over stations of $|\Delta v|$, the absolute difference between the two velocities, per component. It was defined before the freeze and implemented on 12 October 2026, after it.

4.3. Planted and real signals

Gaussian transients have the form

$$s(d,t) = A \exp\left(-\frac{d^2}{2L^2}\right) g(t)\,\mathbf{u}, \tag{4}$$

where d is the distance of a station from the centre, A the amplitude, L the width, $\mathbf{u}$ a horizontal unit vector in a random direction and $g(t)$ the time profile: a raised-cosine rise over D days, a 60-day hold and a raised-cosine fall over D days, after which the transient is zero, so it leaves no permanent offset. s is zero beyond $3L$, and every station in the footprint moves in the same direction. The grid is $A \in \{2, 5, 10\}$ mm, $L \in \{10, 25, 50, 100\}$ km and $D \in \{10, 30, 90\}$ days. Each of the 36 combinations is planted at 20 centres, each at a randomly chosen scored station, giving 720 transients per filter and radius, of which 714–719 have at least one scored station within $2L$. Onsets fall in 300-day slots of the scored period, and the signal is added only on days with data. Many transients share one run of a filter: two share a run only if no station's cleaned series can be changed by both, so the Δ at every station comes from one transient, and fewer transients fit in a run at larger radii.

Fault-slip events are computed from source models taken from published studies and moved to other faults of the same style in the SCEC Community Fault Model 7.0 (Plesch et al., 2007, 2024). Fault surfaces (UTM zone 11, NAD27) are triangulated; a patch of the template's length and depth range

is placed at a random position along a fault of the same style, with dip within $\pm 15^\circ$ of the template's; slip is tapered to zero over the outer 20% of the patch (not at the top when it reaches the surface); and east, north and up displacements at every station are computed from triangular dislocations in an elastic half-space with a Poisson's ratio of 0.25 (Nikkhoo and Walter, 2015). Every template value is a range from the full text of its source, drawn per event. Shallow slow slip follows the 2023 Superstition Hills and Imperial fault events (Materna et al., 2024; Vavra et al., 2024): 8–25 km long, from the surface to 2–4 km depth, 8–45 mm of slip, a slip front moving at 0.4–9 km/day with a local rise of 1–8 days. Afterslip follows the 2010 El Mayor–Cucapah earthquake (Gonzalez-Ortega et al., 2014): 15–50 km long, from 0–2 to 5–12 km depth, 10–40 cm of slip with a logarithmic time function (τ = 17–34 days), followed for 365 days. Slip is permanent after the event. An event is kept only where at least three scored stations move by more than 1 mm, and two events share a run only if no station moves by more than 0.1 mm in both. Twenty events of each template are planted, the same for every filter. Retention is computed as in Eq. (3) over the stations moving by more than 1 mm, and also per station, with the station's distance from the slipping patch and its side of the fault. On the test years the afterslip events move a median of 28.5 scored stations by more than 1 mm, with a median largest displacement of 14.3 mm; the shallow slow-slip events move a median of 3.5 stations, with 4.6 mm. A third template, shallow creep triggered on a neighbouring fault (Ramos et al., 2020; Xu et al., 2020), was dropped because only 1 of 300 random placements moved three or more scored stations by more than 1 mm. The elastic half-space, uniform tapered slip, afterslip without its earthquake and placement on any fault of the right style are the limits of this test.

Real earthquake offsets are every catalogued earthquake step in the scored period at a scored station, with an observed horizontal offset of at least 3 mm and no other catalogued step within 15 days. The offset is the mean position over the 10 days after the earthquake minus that over the 10 days before (at least 5 available days on each side), and the fraction kept is $\langle \text{kept}, \text{observed} \rangle / \langle \text{observed}, \text{observed} \rangle$, with "kept" the same difference measured on the cleaned series. On the test years there are 81 such offsets (median 6.8 mm): 54 from the 5 December 2024 M7.0 Cape Mendocino earthquake, 18 from the 20 December 2022 M6.4 Ferndale earthquake, 5 from an M5.6 event on 24 June 2026 and 2 each from two M5.7 events (9 December 2024 and 14 April 2026; catalogue identifiers nn00888580 and nn00914068).

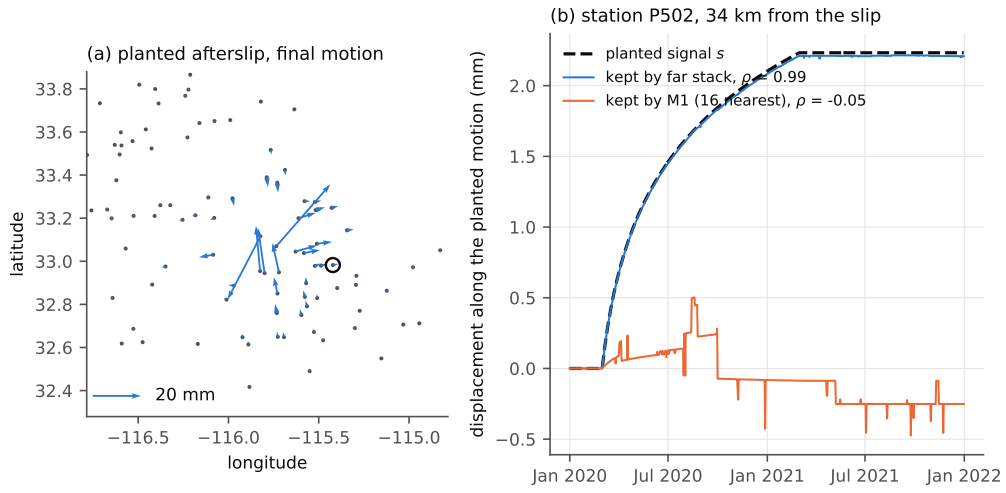


Figure 5: Measuring signal kept, at station P502, the median station among the 17 at 15–40 km from a planted afterslip patch (validation years).

These offsets have no ground truth, since the observed offset includes noise.

The Ridgecrest test uses the real postseismic displacement after the 6 July 2019 mainshock: at the 182 scored stations within 80 km, the horizontal displacement over days 7–365 relative to the mean of days 1–6 is compared with the same quantity in the cleaned series, as $\langle \text{kept}, \text{observed} \rangle / \langle \text{observed}, \text{observed} \rangle$ per station, and the median over stations is reported. These days are excluded from training, and 2019 is in neither the validation nor the test years, so the result is the same for every period. Figure 5 illustrates the measurement of signal kept at one station.

4.4. Protocol

All design decisions, hyper-parameters and model choices were made on the validation years. The method was frozen on 11 October 2026 (repository tag `freeze-2026-10-11`), and the results below are from the test years unless marked otherwise. The test years are not untouched: they were looked at while the scoring rules were being revised on 7 October 2026. The untouched evaluation will be the prospective one, once NGL has published a year of days after the freeze. After the freeze, three additions were made without changing the data, filters or scoring rules: velocity repeatability (defined before the freeze and implemented on 12 October), the split of the bootstrap

comparison by seen and held-out stations (12 October), and the figure code. The attention network was also run at further radii, and further training seeds were run, as allowed by the freeze rules.

Differences in reconstruction error are tested with a paired block bootstrap (`gnssdl bench compare`): each station's leave-one-out fast error is summarised per 30-day block, and blocks (the same for every station) and stations are resampled 1,000 times to give a 95% interval for the difference in the median, run separately for seen stations, held-out stations and all stations. Differences in retention are tested by resampling the planted transients, paired by identifier, for each width. A difference whose interval includes zero is reported as not distinguishable. Trained filters were run with five seeds.

5. Filters

Table 2 lists the filters. They fall into the three families of Section 1; the learned filters belong to the leave-one-out family but are described separately. No leave-one-out filter reads the target's own values on any day. The "days used" column gives the days of the other stations that a prediction for day t reads.

5.1. Target-inclusive filters

C1 is the stack of Wdowinski et al. (1997): the common mode of every station on a day is the unweighted mean of all available stations that day, the station itself included. C3 is sub-regional probabilistic PCA (Tipping and Bishop, 1999), applied as in Gruszczynski et al. (2018) with regions as in Liu et al. (2026): stations are grouped into 1, 4 or 8 regions by k-means on their projected coordinates; in each region and component, a probabilistic PCA with 1, 2 or 3 components is fitted by expectation–maximisation on the quiet training residuals (Section 5.4), normalised per station, with missing days treated as latent; on any day the component scores are estimated from every available station of the region, the target included, and the common mode is the reconstruction. The nine settings are reported as a range, and individually in Table 6. M2-self is the robust weighted median of M2 (Section 5.3) with the target's own value added to the median, as in CMC Imaging (Tian and Shen, 2016; Kreemer and Blewitt, 2021); it reuses M2's tuned settings.

5.2. Distance-excluded filters

The far stack is the unweighted mean of every available station at least R from the target, the target excluded, per day and component. It has no

Table 2: The 17 filter configurations audited, one row per configuration, named as in the results tables. Stations read: the other stations a prediction for the target uses; "nearest" means the nearest stations beyond R. Radii: exclusion radii at which results are reported. Days used: days of other stations read by a prediction for day t (31 d: a 31-day window used only to screen outliers; "all" and 64 d include past and future days). Parameters: learned or tuned parameters at $R = 0$.

Name	Description	Stations read	Radii (km)	Days used	Parameters
Target-inclusive					
C1	Unweighted mean of the whole network	All + target	–	t	0
C3	Sub-regional probabilistic PCA, nine settings (Table 6)	Region + target	–	t	∼3.7k/comp.
M2-self	Robust weighted median of the neighbours and the target	16 nearest + target	–	t (31 d)	2
Distance-excluded					
Far stack	Unweighted mean of every station beyond R	All beyond R	0, 25, 50, 100, 200, 400, 600	t	0
D1	Far stack at $R = 400$ km, computed separately	All beyond 400 km	400	t	0
Leave-one-out over nearest neighbours					
M1	Distance-weighted mean, $\ell = 50$ km at $R = 0$	16 nearest	0, 10, 25, 50, 100, 200, 400	t	1
M1, 64	M1 over 64 neighbours, $\ell = 50$ km	64 nearest	0, 100, 400	t	1
M1, 256	M1 over 256 neighbours, $\ell = 100$ km	256 nearest	0, 100, 400	t	1
M2	Robust weighted median after outlier screening	16 nearest	0	t (31 d)	2
M3a	Ridge regression, one weight vector per station	16 nearest	0	t	65/station
M3b	Ridge regression pooled over stations, with neighbour geometry	16 nearest	0	t	257
M4a	M3a per period band (under 7, 7–60, 60–400, over 400 days)	16 nearest	0	All	4× M3a
M4b	M3b per period band	16 nearest	0	All	4× M3b
Learned					
M3k	Learned kernel: neighbour weight from distance, azimuth and noise level	16 nearest	0, 100, 400	t	1,411
M3k, 256	M3k over 256 neighbours	256 nearest	0, 100, 400	t	1,411
M5	Space–time attention, correction to the far stack	32 nearest + far stack	0, 10, 25, 50, 100	64 d	212,048
M5-aug	M5 trained with planted transients in its inputs	As M5	0, 25	64 d	212,048

parameters and is run at $R = 0$, 25, 50, 100, 200, 400 and 600 km. D1 is the far stack at $R = 400$ km, the distance of Bachelot et al. (2025), computed separately; its results agree with those of the far stack at 400 km. Bachelot et al. (2025) use the eight nearest stations beyond 400 km in a graph neural network and compare it with a stack of the same stations; D1 uses every station beyond 400 km.

5.3. Leave-one-out filters over nearest neighbours

These filters predict the target from its K nearest stations beyond R, using only day t (M1–M3) or the neighbours' whole series (M4). Their settings were chosen on the validation years by reconstruction error.

M1 is the weighted mean $\hat{r}_i(t) = \sum_j w_{ij} a_j(t) r_j(t) / \sum_j w_{ij} a_j(t)$ over the 16 nearest stations, with $w_{ij} = \exp(-d_{ij}^2/2\ell^2)/\bar{\sigma}_j^2$, $a_j(t)$ the availability and $\bar{\sigma}_j$ the station's median formal uncertainty over the training years, per component. The length scale ℓ was chosen from 10 km to infinity (a plain mean): 50 km at $R = 0$ for 16 and 64 neighbours, and 100 km for 256. M1 with 64 and 256 neighbours forms the neighbour-count experiment.

M2 follows Kreemer and Blewitt (2021): a neighbour's value on day t is dropped if it differs from that neighbour's own 31-day running median by more than k times its robust scale; the rest are combined by a weighted median with M1's weights, interpolated as in Kreemer and Blewitt (2021). With fewer than three neighbours left it falls back to M1. The tuned values at $R = 0$ are $\ell = 200$ km and $k = 5$.

M3 is ridge regression on the three components of the 16 neighbours, an availability bit per neighbour and an intercept, in units of each station's noise scale. Missing neighbours are set to zero and the remaining values rescaled by K divided by the number available, so that gaps do not bias the fit. Rows are replicated four times with neighbours dropped at random. M3a fits one weight vector per station and component (65 features; $\lambda = 0.1$ in units of the trace of the Gram matrix per feature); the 178 stations without enough training rows, the held-out stations among them, use M3b. M3b fits one weight vector per component for all stations, adding each neighbour slot's log distance and azimuth and their products with the slot's residuals (257 features; $\lambda = 0.01$).

M4 applies M3 separately to four period bands that add up exactly to each series (under 7, 7–60, 60–400 and over 400 days), obtained with differences of zero-phase Gaussian smoothers, and sums the four predictions; λ is chosen per band. Before filtering, each series' gaps are filled by linear interpolation

of its own days, and a neighbour missing on a day contributes zero with its availability bit off. The band filters read the neighbours' past and future days. M4a is per station and M4b pooled, as M3a and M3b.

5.4. Learned filters

M3k and M5 are trained on quiet residuals: the residuals minus the earthquake offsets estimated in the training-years fit, on training days, outside the Ridgecrest exclusion and outside ±30 days of catalogued M ≥ 6 earthquakes, at stations that are not held out. The loss is the squared error in units of each station's robust noise scale. Training stops early on the validation reconstruction error, and each model was trained with five seeds; the reported results are those of seed 0.

M3k keeps M1's form and learns its kernel: $w_{ijc} = \mathrm{softplus}(f_\theta(g_{ij}, q_j))_c$, where g_{ij} is the neighbour's log distance and the sine and cosine of its azimuth from the target, q_j its log robust noise scale per component and record completeness, and f_θ a multilayer perceptron with two hidden layers of 32 units (1,411 parameters). The weights depend on where a neighbour is and what kind of station it is, never on its values, so once trained M3k is a fixed weighted mean of day t. It was trained over 16 and over 256 neighbours at every radius; at $R = 0$ the best epoch was 14 of 20 (16 neighbours) and 4 of 10 (256).

M5 is a space–time attention network. A sample is a target, its 32 nearest stations beyond R, chosen for each 64-day window among the 64 nearest by availability, and one summary node, the far stack. Each node has 17 input channels per day: residual, log formal uncertainty, availability, a flag for cells hidden by a training mask, the daily increment $r(t) - r(t-1)$ (zero unless both days exist), the station's log noise scale and record completeness, and the day of year (sine and cosine), all in units of the station's noise scale. A convolution cuts each node's window into 4-day tokens; three blocks then alternate attention over a node's own tokens (time) and attention over the nodes at the same token (space), the latter biased by each node's log distance, azimuth and type relative to the target ($d = 64$, four heads). The output is a correction added to the far stack, so training starts from the far stack. The target is hidden for the whole window, so predictions never read it. Training used batches of 64 samples, a learning rate of 10^{-3} with 500 warm-up steps, weight decay 10^{-4}, dropout 0.1, neighbour dropout 0.3, and early stopping every 500 steps (best step 5,000 at $R = 0$). Prediction uses 64-day windows overlapping by half, combined with a triangular taper, so a day's prediction

reads neighbours' days before and after it. M5 was trained at $R = 0$, 10, 25, 50 and 100 km.

M5-aug is the same network trained with planted transients. In half of the training samples a synthetic field is added to the inputs of every node and to the far-stack node as it would enter that mean, but not to the targets, so copying it into a prediction increases the loss. The field is centred at a random point within $R + 300$ km of the target, with a width of 5–150 km, an amplitude of 1–15 times the station's noise scale, a duration of 5–60 days with a raised-cosine rise and fall, and a random horizontal direction; half are Gaussian blobs and half dipoles, whose sign flips across a random line through the centre. The generator uses its own seed and shapes and runs on training days only, so it is separate from the evaluation transients. M5-aug was trained at $R = 0$ and 25 km (best step 1,500 at $R = 0$).

5.5. Implementation and use of AI tools

The `gnssdl` package, its tests (120 automated tests, including the leak checks), the design documents and the figure code were written by Claude Opus 5.5 (Anthropic) through Claude Code, under the author's direction, between 6 and 13 October 2026. The research questions are the author's. The code is openly available at `https://github.com/hogib/gnssdl` under the MIT licence; per-model results, logs and checkpoints are kept in `data/runs/<model>/`. This report, including its Discussion and Conclusions, was drafted by the same tool from the repository and its results files.

6. Results

Table 3 gives every measure for every configuration of Table 2 at $R = 0$ (D1 at 400 km) on the test years, with the nine C3 settings listed separately and the unfiltered residuals as a reference. The subsections give the supporting numbers. Validation-year values are marked (val).

6.1. The trade-off between noise removed and signal kept

Figure 6 plots noise removed against retention of 100-km transients for every filter, with the radius sweeps as lines. Table 4 gives the sweeps. For the far stack, from $R = 0$ to 400 km, retention of 100-km transients rises from 0.83 to 1.00, spurious signal from 0.12 to 0.25 and reconstruction error from 0.566 to 0.585, and noise removed falls from 22.7% to 20.5%; at 600 km

Table 3: All measures for every filter configuration, $R = 0$ (D1: $R = 400$ km), test years (2022 to September 2026). Error: leave-one-out reconstruction error (lower is better; target-inclusive filters are not scored on it). Noise: noise removed, Eq. (2). Gaussian transients: median retention (kept) and median spurious signal by width L. Afterslip: whole-event retention and spurious signal, and per-station retention at 15–40 km and beyond 40 km from the patch. Shallow slow slip: whole-event retention and spurious signal. Offsets: fraction of the 81 real earthquake offsets kept. Ridgecrest: postseismic displacement kept. $|\Delta v|$: median velocity difference between 2022–2023 and 2024–2025 (mm/yr) at 590 stations. –: not applicable or not run (M1 over 64 neighbours was not run on the fault-slip events, the real offsets or velocity repeatability). C3 settings: k regions, p components.

			Gaussian, kept				Gaussian, spurious				Afterslip				Shallow slow slip				$\|\Delta v\|$ (mm/yr)		
Filter	Error	Noise	10	25	50	100 km	10	25	50	100 km	Kept	Spur.	15–40	>40 km	Kept	Spur.	Offsets	Ridgecrest	E	N	U
No filter	0.753	0.0%	1.00	1.00	1.00	1.00	0.00	0.00	0.00	0.00	1.00	0.00	1.00	1.00	1.00	0.00	1.00	1.00	1.98	0.82	4.59
C1	–	22.8%	1.00	0.98	0.94	0.83	0.00	0.01	0.04	0.12	1.00	0.00	1.00	0.98	1.00	0.00	0.88	0.91	0.83	1.53	2.41
C3, $k = 1$, $p = 1$	–	22.7%	1.00	0.99	0.95	0.85	0.04	0.16	0.60	1.58	1.00	0.03	1.00	0.98	1.00	0.01	0.87	0.92	1.33	1.21	2.16
C3, $k = 1$, $p = 2$	–	23.1%	1.00	0.98	0.94	0.82	0.02	0.09	0.27	0.58	1.00	0.02	1.00	0.98	1.00	0.01	0.87	0.92	0.94	1.38	1.77
C3, $k = 1$, $p = 3$	–	21.9%	0.99	0.98	0.93	0.81	0.04	0.12	0.27	0.57	1.00	0.04	1.00	0.98	1.00	0.02	0.87	0.92	0.86	1.24	1.78
C3, $k = 4$, $p = 1$	–	19.8%	0.99	0.95	0.83	0.58	0.02	0.09	0.24	0.39	1.00	0.01	0.99	0.97	1.00	0.01	0.82	0.94	1.24	0.79	2.10
C3, $k = 4$, $p = 2$	–	19.5%	0.99	0.94	0.81	0.56	0.06	0.17	0.35	0.43	1.00	0.06	0.99	0.98	1.00	0.05	0.78	0.94	1.23	1.07	2.11
C3, $k = 4$, $p = 3$	–	19.0%	0.98	0.93	0.78	0.54	0.07	0.17	0.40	0.46	1.00	0.07	0.99	0.96	0.99	0.07	0.78	0.93	1.43	0.97	2.11
C3, $k = 8$, $p = 1$	–	16.2%	0.98	0.90	0.71	0.35	0.03	0.12	0.33	0.24	1.00	0.02	0.99	0.90	0.99	0.01	0.68	0.84	1.13	1.41	1.62
C3, $k = 8$, $p = 2$	–	19.1%	0.97	0.88	0.66	0.32	0.07	0.18	0.40	0.28	0.99	0.06	0.97	0.89	0.99	0.04	0.67	0.83	1.13	0.96	1.82
C3, $k = 8$, $p = 3$	–	19.1%	0.96	0.86	0.65	0.31	0.09	0.23	0.44	0.29	0.99	0.07	0.97	0.89	0.98	0.05	0.69	0.79	1.14	1.12	2.07
M2-self	–	19.3%	0.75	0.24	0.03	0.00	0.43	0.31	0.17	0.11	0.79	0.04	0.18	−0.04	0.98	0.21	0.14	0.33	0.81	0.87	1.48
Far stack	0.566	22.7%	1.00	0.98	0.94	0.83	0.00	0.01	0.04	0.12	1.00	0.00	1.00	0.98	1.00	0.00	0.88	0.92	0.83	1.53	2.42
D1	0.585	20.5%	1.00	1.00	1.00	1.00	0.01	0.04	0.11	0.25	1.00	0.01	1.00	1.00	1.00	0.00	0.91	0.91	0.85	2.50	2.56
M1	0.621	14.1%	0.77	0.25	0.05	0.00	0.20	0.12	0.06	0.02	0.87	0.01	0.11	−0.05	1.00	0.09	0.09	0.31	1.15	1.17	2.27
M1, 64	0.596	18.0%	0.90	0.53	0.18	0.03	0.11	0.18	0.10	0.04	–	–	–	–	–	–	–	0.58	–	–	–
M1, 256	0.606	17.2%	0.97	0.84	0.52	0.20	0.03	0.13	0.22	0.11	1.00	0.02	0.96	0.75	1.00	0.01	0.45	0.84	1.03	1.31	2.18
M2	0.638	12.2%	0.82	0.28	0.04	0.00	0.50	0.33	0.19	0.12	0.86	0.04	0.20	−0.04	0.99	0.27	0.12	0.37	0.90	1.04	1.70
M3a	0.603	16.6%	0.83	0.39	0.21	0.16	0.20	0.13	0.06	0.04	0.91	0.01	0.23	0.06	0.99	0.09	0.20	0.50	1.06	1.06	2.03
M3b	0.599	15.6%	0.78	0.30	0.11	0.07	0.21	0.16	0.07	0.04	0.87	0.01	0.24	0.08	0.99	0.07	0.21	0.50	1.10	1.04	2.31
M4a	0.582	19.5%	0.88	0.52	0.39	0.34	0.17	0.12	0.06	0.04	0.94	0.01	0.54	0.41	1.00	0.09	0.23	0.85	1.24	0.98	2.41
M4b	0.598	15.9%	0.78	0.33	0.16	0.11	0.20	0.14	0.07	0.04	0.88	0.01	0.33	0.19	0.99	0.08	0.20	0.50	1.16	1.05	2.45
M3k	0.595	18.0%	0.77	0.25	0.04	0.01	0.22	0.15	0.06	0.03	0.86	0.01	0.13	−0.05	0.97	0.08	0.22	0.36	0.87	1.01	1.96
M3k, 256	0.562	22.7%	0.88	0.70	0.45	0.20	0.05	0.09	0.22	0.18	0.94	0.02	0.72	0.47	0.97	0.02	0.61	0.76	0.87	1.04	1.59
M5	0.563	22.5%	0.95	0.83	0.72	0.59	0.08	0.09	0.09	0.16	0.98	0.01	0.90	0.77	0.99	0.03	0.78	0.91	0.73	1.48	2.17
M5-aug	0.565	22.6%	1.00	0.98	0.93	0.82	0.02	0.03	0.07	0.16	1.00	0.01	0.99	0.97	1.00	0.01	0.88	0.91	0.85	1.47	2.17

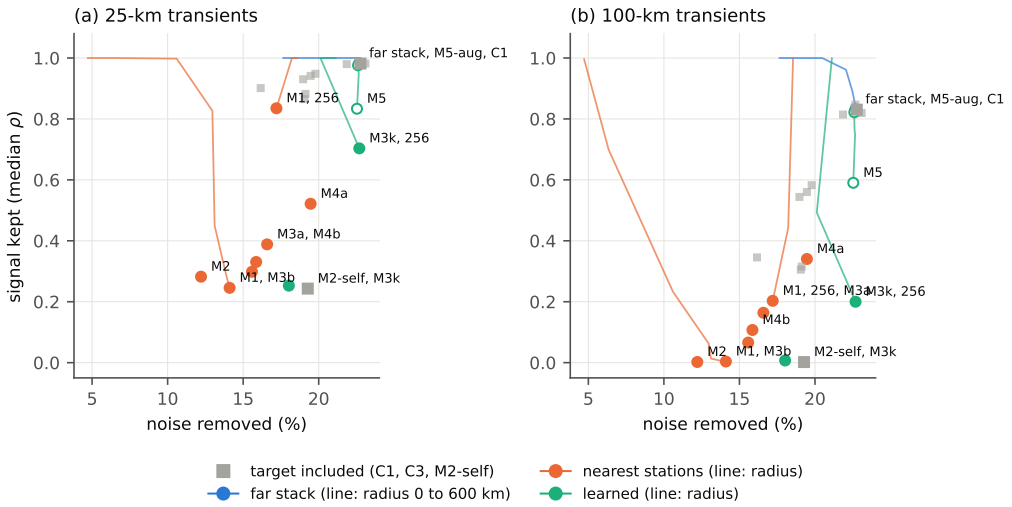


Figure 6: Noise removed against retention of 100-km Gaussian transients, test years. Points: filters at $R = 0$; lines: radius sweeps.

noise removed is 17.7% and spurious signal 0.33. For M1, retention of 100-km transients rises from 0.00 at $R = 0$ to 1.00 at 400 km, spurious signal from 0.02 to 0.78, and noise removed falls from 14.1% to 4.7%. For M5, from $R = 0$ to 100 km, retention of 100-km transients rises from 0.59 to 0.83 and noise removed stays between 22.5% and 22.7%; spurious signal is 0.14–0.16 throughout. M5-aug at $R = 25$ km keeps 0.98 and 0.83 of 25- and 100-km transients and removes 22.6%.

6.2. Number of stations versus distance

Figure 7 compares filters that differ in the number of stations they average. At $R = 0$ on the test years, the reconstruction error is 0.621 for M1 over 16 neighbours, 0.596 over 64, 0.606 over 256 and 0.566 for the far stack, which averages all 1,250 other stations; on the validation years the same four values are 0.531, 0.511, 0.507 and 0.509 (val). Noise removed on the test years is 14.1%, 18.0%, 17.2% and 22.7%. Retention of 25-km transients is 0.25, 0.53, 0.84 and 0.98, and of 100-km transients 0.00, 0.03, 0.20 and 0.83. At $R = 100$ km the reconstruction errors are 0.660 (16), 0.601 (64), 0.599 (256) and 0.569 (far stack).

Regression on the same 16 neighbours gives reconstruction errors of 0.603 (M3a), 0.599 (M3b), 0.582 (M4a) and 0.598 (M4b), against 0.621 for M1;

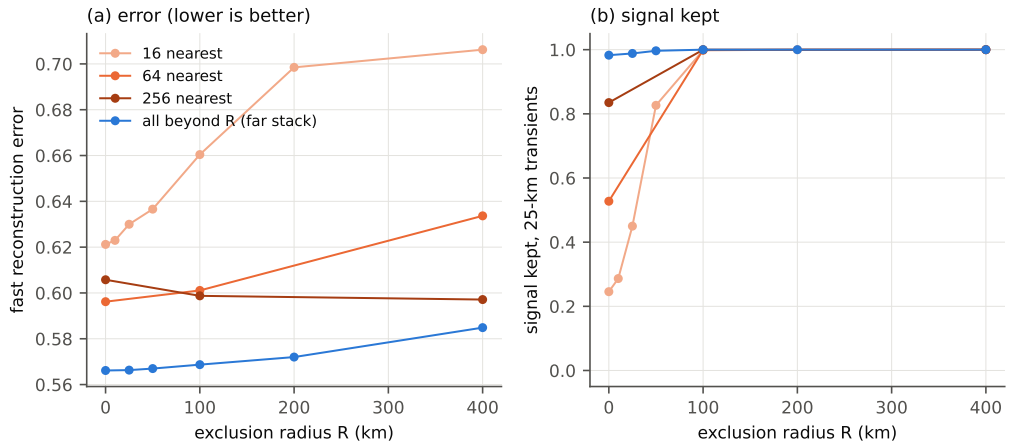


Figure 7: Reconstruction error (a) and retention of 25-km transients (b) against exclusion radius for M1 over 16, 64 and 256 neighbours and the far stack, test years.

noise removed is 16.6%, 15.6%, 19.5% and 15.9%. Retention of 25- and 100-km transients is 0.39 and 0.16 (M3a), 0.30 and 0.07 (M3b), 0.52 and 0.34 (M4a) and 0.33 and 0.11 (M4b), against 0.25 and 0.00 for M1. On the validation years the per-station regressions had lower errors than the pooled ones (M3a 0.527, M3b 0.534; M4a 0.517, M4b 0.532; val); on the test years M3a's error is higher than M3b's.

6.3. Retention by width, amplitude and duration

Table 5 gives retention for each width. Across the three amplitudes (2, 5 and 10 mm), median retention of a given width changes by at most 0.15 for any filter (M3k at 25 km: 0.20, 0.35 and 0.25); for M1 at 25 km it is 0.20, 0.30 and 0.25, for the far stack at 100 km 0.85, 0.82 and 0.83, and for M5 at 100 km 0.57, 0.58 and 0.63. Across the three durations (10, 30 and 90 days) it changes by at most 0.20 (C3 with four regions and three components at 100 km: 0.54, 0.41 and 0.61); for the far stack at 100 km it is 0.83, 0.81 and 0.86, and for M4a at 100 km 0.29, 0.33 and 0.41. Spurious signal for 10-km transients is 0.20 for M1, 0.50 for M2 and 0.43 for M2-self, against 0.00 for the far stack; for 100-km transients it is 0.02, 0.12, 0.11 and 0.12 (Table 5).

6.4. Sub-regional PCA settings

Table 6 gives the nine C3 settings. With one region, retention of 100-km transients is 0.81–0.85 and spurious signal 0.57–1.58, the largest value being

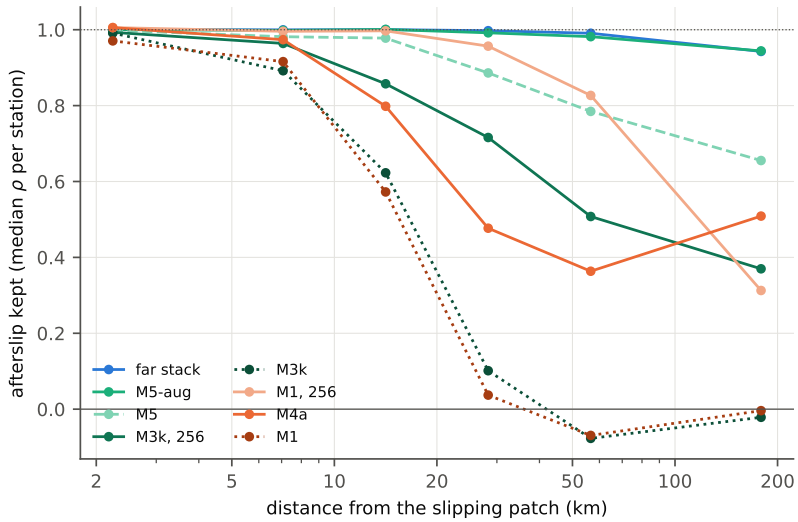


Figure 8: Afterslip kept per station against distance from the slipping patch, $R = 0$, test years.

for one component. With eight regions, retention of 100-km transients is 0.31–0.35 and spurious signal 0.24–0.29.

6.5. Realistic transients

Figure 8 and Table 7 give afterslip retention per station against distance from the slipping patch. Within 5 km every filter keeps 0.94–1.01. At 15–40 km retention is 0.11 (M1), 0.13 (M3k), 0.20 (M2), 0.18 (M2-self), 0.23 (M3a), 0.24 (M3b), 0.33 (M4b), 0.54 (M4a), 0.72 (M3k-256), 0.90 (M5), 0.96 (M1-256), 0.99 (M5-aug) and 1.00 (far stack, C1, D1). Beyond 40 km it is −0.05 (M1), −0.05 (M3k), 0.41 (M4a), 0.47 (M3k-256), 0.75 (M1-256), 0.77 (M5), 0.97 (M5-aug) and 0.98 (far stack). Within 15 km of the patch, the two sides of the fault differ by up to 0.11 (M3a 0.99 and 0.88; M3k 0.94 and 0.83; M1 0.95 and 0.88), and by at most 0.02 for the far stack, C1, M5 and M5-aug. Retention over the whole event, over all stations moving by more than 1 mm, is 0.79–1.00 for every filter (lowest M2-self 0.79, M2 0.86, M3k 0.86, M1 0.87).

The shallow slow-slip events are kept by every filter (0.97–1.00 over the whole event). The spurious signal they create, as a fraction of the event's largest displacement, is 0.27 for M2, 0.21 for M2-self, 0.07–0.09 for M1, M3 and M4, 0.08 for M3k, 0.03 for M5 and 0.00 for the far stack and C1 (Table 7).

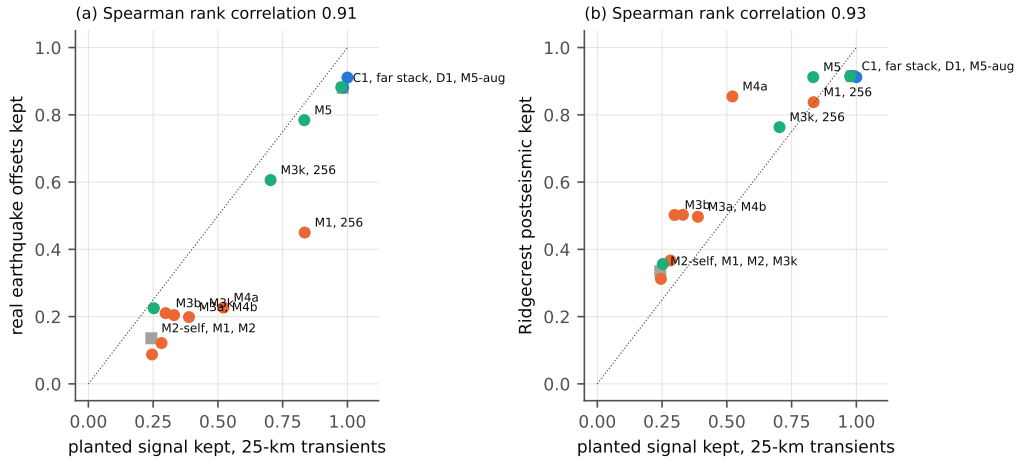


Figure 9: Real earthquake offsets (a) and Ridgecrest postseismic displacement (b) kept, against retention of 25-km Gaussian transients, $R = 0$, test years.

6.6. Real earthquake offsets and Ridgecrest

The fraction of the 81 real offsets kept is 0.88 for the far stack, C1 and M5-aug, 0.91 for D1, 0.78 for M5, 0.61 for M3k-256, 0.45 for M1-256, 0.20–0.23 for M3, M4 and M3k, 0.14 for M2-self, 0.12 for M2 and 0.09 for M1 (Table 3). On the validation years, with 37 offsets mainly from the 2020 Monte Cristo Range, 2021 Antelope Valley and 2021 Petrolia earthquakes, the fraction kept was 0.98 for the far stack, 0.99 for D1, 0.98 for M5-aug, 0.88 for M5 and 0.72 for M1 (val). Of the Ridgecrest postseismic displacement, the far stack keeps 0.92, C1, D1, M5 and M5-aug 0.91, M4a 0.85, M1-256 0.84, M3k-256 0.76, M3a, M3b and M4b 0.50, M2 0.37, M3k 0.36, M2-self 0.33 and M1 0.31. Figure 9 plots both against retention of 25-km Gaussian transients; the Spearman rank correlation over the fifteen filters is 0.91 for the offsets and 0.93 for Ridgecrest (0.94 and 0.92 on the validation years).

6.7. Learned models

Over 16 neighbours, M3k has a reconstruction error of 0.595 against 0.621 for M1, removes 18.0% against 14.1%, and keeps 0.25 and 0.01 of 25- and 100-km transients against 0.25 and 0.00. On the validation years the bootstrap interval of the error difference between M3k and M1 was 0.008–0.021 (val); this comparison was not rerun on the test years. Over 256 neighbours, M3k

has an error of 0.562 against 0.606 for M1-256, removes 22.7% against 17.2%, and keeps 0.70 and 0.20 against 0.84 and 0.20.

Table 8 gives the bootstrap comparisons with the far stack on the test years. For M5, M5-aug and M3k-256, the 95% interval of the error difference includes zero on seen stations, on held-out stations and on all stations. On held-out stations the medians are 0.5569 (far stack), 0.5470 (M5), 0.5543 (M5-aug) and 0.5522 (M3k-256). The retention of each of the three is lower than that of the far stack at every width, with intervals that exclude zero except for M5-aug at 10 km, whose interval reaches zero; for M5-aug the differences are 0.001–0.009, for M5 0.05–0.24 and for M3k-256 0.12–0.63. The bootstrap medians are computed from 30-day block summaries and differ slightly from the errors in Table 3.

Across five seeds, the mask-based `scatter` scores at $R = 0$ (not the reconstruction error) range over 0.509–0.510 (val) and 0.618–0.620 (test) for M3k, 0.478–0.483 and 0.558–0.567 for M3k-256, 0.479–0.481 and 0.548–0.553 for M5, and 0.482–0.485 and 0.551–0.553 for M5-aug.

On the validation years, M3k-256 had the lowest reconstruction error of all filters (0.499), and its difference from the far stack (0.509) had a bootstrap interval of 0.001–0.015 (val). M5 and M5-aug had errors of 0.504 and 0.508 (val).

6.8. Velocity repeatability

Table 9 gives the median $|\Delta v|$ between 2022–2023 and 2024–2025 at the 590 scored stations with a result. Without filtering it is 1.98, 0.82 and 4.59 mm/yr in east, north and up. Every filter lowers the east and up values. The north value rises with every filter except one C3 setting (four regions, one component: 0.79), to 1.53 for the far stack and C1, 2.50 for D1, 1.48 for M5 and 1.17 for M1. The lowest east value is that of M5 (0.73) and the lowest up value that of M2-self (1.48).

7. Discussion

This section and the next are the drafting tool's interpretation of the results, written so that the study can be read as a whole. They are one reading of the numbers, not the author's, and the paper's discussion is to be written independently.

7.1. What sets the signal a filter removes

The results are consistent with the hypothesis of the proposal: the fraction of a transient that a filter removes depends mainly on how much of the filter's input lies inside the transient. For a linear filter, Eq. (3) gives $\rho = 1 - \langle P(s), s\rangle / \langle s, s\rangle$, so the loss is the part of the transient that the filter's own prediction reproduces. For the far stack at $R = 0$, the prediction is the network mean of s, and a Gaussian transient contributes to that mean in proportion to the share of stations it covers. Computing this share from the station positions alone, ignoring data gaps, gives median retentions of 1.00, 0.985, 0.95 and 0.85 for widths of 10, 25, 50 and 100 km, against measured values of 1.00, 0.98, 0.94 and 0.83. The far stack therefore loses a 100-km transient for a simple geometric reason: such a transient covers a noticeable fraction of a network that is itself only a few hundred kilometres across.

For the nearest-neighbour filters the same reasoning works the other way. The 16th neighbour lies at a median of 40 km, so a 25-km transient covers most of a station's neighbours and a 100-km transient covers all of them; M1 keeps 0.25 and 0.00 of these. Increasing the exclusion radius moves the neighbours outside the transient, and retention rises when R becomes comparable with $2L$ to $3L$: M1 keeps 0.83 of 25-km transients at $R = 50$ km, 0.73 of 50-km transients at $R = 100$ km and 0.70 of 100-km transients at $R = 200$ km. The price is that the remaining neighbours are farther away and fewer stations share their noise with the target, so noise removed falls from 14.1% to 4.7%, and spurious signal rises, because a transient now reaches stations whose neighbours are inside it while they themselves are not.

The exclusion radius is therefore not a parameter with an optimum but a choice of which transients to protect. For the far stack, raising R from 0 to 200 km raises retention of 100-km transients from 0.83 to 0.96 at a cost of 0.6 points of noise removed and 0.006 in reconstruction error; at 400 km, all transients are kept, but noise removed falls by 2.2 points and spurious signal doubles. Which point is preferable depends on the transients a study is looking for.

7.2. How many stations, and how far

On the validation years the weighted mean of the 256 nearest stations and the far stack had the same reconstruction error, which suggested that the number of stations averaged, not their distance, is what reduces the error. The test years do not support that reading in its simple form: the far stack's error is lower by 0.04, and the 64-neighbour mean has a lower error than the

256-neighbour one. The two filters differ in more than the number of stations, since M1 weights its neighbours by distance with a length scale tuned on the validation years, and the comparison cannot separate the two effects. What holds in both periods is that averaging more stations keeps more signal (25-km retention 0.25, 0.53, 0.84 and 0.98 for 16, 64, 256 and all stations), and that no filter over a subset of nearby stations removed more noise than the mean over all of them on the test years.

7.3. Local filters and the far field

The fault-slip events show a pattern that the Gaussian transients cannot: the nearest-neighbour filters keep afterslip close to the fault and remove it farther away. Within 5 km of the patch, every filter keeps 0.94–1.01; at 15–40 km, M1 keeps 0.11 and M3k 0.13, while the far stack keeps 1.00. Near the fault the displacement varies over distances shorter than the spacing of the neighbours, and on a strike-slip fault the two sides move in opposite directions, so the neighbours' mean does not reproduce the motion at the target. Far from the fault the displacement field is smooth over the scale of the neighbourhood, and a neighbour average reproduces it, as it does for a wide Gaussian transient. The far field of afterslip and slow slip constrains the depth and moment of the slip, so a local filter removes the part of the signal that a source inversion relies on. The Gaussian transients are in this sense pessimistic near faults and optimistic far from them, and the whole-event retention, dominated by the near-fault stations with the largest motion (0.79–1.00 for every filter), hides the difference.

7.4. Learned filters

Learning helps a filter that is restricted to the same inputs: the learned kernel has a lower error than the fixed Gaussian kernel over 16 neighbours (0.595 against 0.621) and over 256 (0.562 against 0.606), and removes more noise, at the same or lower retention. None of the learned filters, however, has a reconstruction error distinguishable from that of the far stack on the test years, on seen or on held-out stations, and every one keeps less planted signal. The attention network is built as the far stack plus a learned correction; without planted transients in training, the correction removes part of every coherent transient (100-km retention 0.59 against 0.83) without a measurable reduction in error, and with them (M5-aug), it keeps signal almost as well as the far stack and removes the same noise. On the validation years, the learned kernel over 256 neighbours had a lower error than the far stack with

an interval excluding zero; that this advantage disappeared on the test years suggests that part of what was learned was specific to the validation period, or that the test years, which contain a different mix of noise and earthquakes, do not reward it. On this network and with daily data, the far stack is the baseline the learned filters did not beat, and the measure that separates them from it is signal kept rather than noise removed.

7.5. Target-inclusive filters and the measures

The target-inclusive filters illustrate why scatter reduction alone misleads. M2-self is M2 with the target added to the median: it removes 19.3% of the scatter against 12.2% for M2, while keeping the same signal (0.24 against 0.28 at 25 km) and real offsets (0.14 against 0.12). The extra noise removed is part of the station's own noise, which a filter that reads the target can subtract and a leave-one-out filter cannot; it says nothing about the common mode. C1, the network mean with the target included, behaves like the far stack at $R = 0$ on every measure, since one station out of 1,251 contributes little to the mean. C3 shows a different cost: with one region and one component its spurious signal is 1.58 times the planted amplitude for 100-km transients, because a single component fitted to the whole network spreads part of any large transient over every station; more regions reduce the spurious signal but remove more of the transient. Velocity repeatability rewards the same behaviour as scatter reduction: M2-self has the smallest vertical $|\Delta v|$, and a filter that absorbs real changes in velocity improves this measure. Reconstruction error did not predict retention either: M3k has a lower error than M1 and M2 and keeps as little. Retention and spurious signal are therefore not refinements of the noise measures but independent axes, and a comparison of filters that omits them can rank a filter that removes signal above one that does not.

7.6. Relation to Bachelot et al.

The far stack at 400 km (D1) keeps every planted transient and all of the afterslip, consistent with the choice of Bachelot et al. (2025) to exclude nearby stations, and the far stack performing as well as the learned filters here parallels their finding that a stack of distant stations performed comparably to their network. The audit adds what the fixed distance costs: compared with the far stack at $R = 0$, D1 removes 2.2 points less of the scatter and creates twice the spurious signal for 100-km transients, while the gain in retention is mostly for transients of 50 km and wider (0.94 to 1.00 at 50 km

and 0.83 to 1.00 at 100 km, against 0.98 to 1.00 at 25 km). On this network, a smaller radius keeps transients up to 50 km almost completely (0.96 at $R = 50$ km).

7.7. Limitations

The audit uses one dense network in a strike-slip setting; in a sparse network, such as Türkiye's, where the median distance to the nearest NGL station is about 60 km, every filter's neighbours are farther away and the trade-off shifts. The fault-slip events use an elastic half-space, uniform tapered slip and two templates. The real offsets have no ground truth, and on the test years two thirds of them come from one earthquake. Only daily solutions are used. The test years were looked at while the scoring rules were revised, and several comparisons differ between the validation and test years, so the prospective evaluation is needed before any of the smaller differences reported here can be relied on. Velocity repeatability in the north component worsens with almost every filter for reasons that have not been examined.

8. Conclusions

This section is the drafting tool's summary of its own interpretation (Section 7).

1. The signal a common-mode filter removes is set by how much of its input lies inside the transient. Filters built on the 16 nearest stations remove most of any transient wider than about 25 km; the mean of all stations loses only the share of a transient that enters the network mean, 0.17 of a 100-km transient on this network.
2. The exclusion radius trades signal kept against noise removed and spurious signal; there is no radius that is best on every measure, and the choice depends on the transients of interest.
3. On realistic fault slip, local filters keep the near-fault signal and remove the far field, which Gaussian transients and whole-event averages do not show.
4. On the test years, no learned filter reduced the reconstruction error below that of the far stack, and each kept less signal; training with planted transients brought the attention network's signal kept up to, not beyond, that of the far stack.

5. Scatter reduction, velocity repeatability and reconstruction error all reward filters that remove signal or the station's own noise; signal kept and spurious signal need to be reported alongside them.

Data and code availability

The NGL daily IGS20 time series are available from the Nevada Geodetic Laboratory (Nevada Geodetic Laboratory, 2026; Blewitt et al., 2018), and the SCEC Community Fault Model 7.0 from its Zenodo record (Plesch et al., 2024). The code is at `https://github.com/hogib/gnssdl` (MIT licence); the frozen method is tagged `freeze-2026-10-11`. Results files are in `data/runs/<model>/` (not tracked).

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Table 4: Radius sweeps, test years. Error: leave-one-out reconstruction error (computed for the far stack, M1 and, at $R = 0$, the learned filters). Spurious: 100-km transients.

Filter	R (km)	Error	Noise removed	Kept 25 / 100 km	Spurious	Offsets	Ridgecrest
Far stack	0	0.566	22.7%	0.98 / 0.83	0.12	0.88	0.92
	25	0.566	22.7%	0.99 / 0.84	0.12	–	0.92
	50	0.567	22.6%	1.00 / 0.86	0.13	–	0.92
	100	0.569	22.5%	1.00 / 0.89	0.14	0.88	0.92
	200	0.572	22.1%	1.00 / 0.96	0.16	–	0.93
	400	0.585	20.5%	1.00 / 1.00	0.25	0.91	0.91
	600	0.604	17.7%	1.00 / 1.00	0.33	–	0.88
M1	0	0.621	14.1%	0.25 / 0.00	0.02	0.09	0.31
	10	0.623	13.9%	0.29 / 0.00	0.03	–	0.44
	25	0.630	13.1%	0.45 / 0.01	0.03	–	0.58
	50	0.637	13.0%	0.83 / 0.06	0.05	–	0.74
	100	0.660	10.6%	1.00 / 0.23	0.10	0.78	0.94
	200	0.699	6.3%	1.00 / 0.70	0.41	–	0.94
	400	0.706	4.7%	1.00 / 1.00	0.78	0.92	0.90
M1, 64	0	0.596	18.0%	0.53 / 0.03	0.04	–	0.58
	100	0.601	17.8%	1.00 / 0.33	0.15	–	0.96
	400	0.634	14.2%	1.00 / 1.00	0.62	–	0.90
M1, 256	0	0.606	17.2%	0.84 / 0.20	0.11	0.45	0.84
	100	0.599	18.2%	1.00 / 0.44	0.19	0.82	0.92
	400	0.597	18.5%	1.00 / 1.00	0.44	0.90	0.93
M3k	0	0.595	18.0%	0.25 / 0.01	0.03	0.22	0.36
	100	–	16.2%	1.00 / 0.24	0.10	0.77	0.99
	400	–	12.2%	1.00 / 1.00	0.79	0.92	0.93
M3k, 256	0	0.562	22.7%	0.70 / 0.20	0.18	0.61	0.76
	100	–	20.1%	1.00 / 0.49	0.27	0.79	0.94
	400	–	21.1%	1.00 / 1.00	0.48	0.90	0.93
M5	0	0.563	22.5%	0.83 / 0.59	0.16	0.78	0.91
	10	–	22.7%	0.95 / 0.75	0.14	0.84	0.91
	25	–	22.6%	0.96 / 0.76	0.14	0.84	0.91
	50	–	22.6%	0.99 / 0.81	0.14	0.86	0.90
	100	–	22.5%	1.00 / 0.83	0.15	0.88	0.91
M5-aug	0	0.565	22.6%	0.98 / 0.82	0.16	0.88	0.91
	25	–	22.6%	0.98 / 0.83	0.17	0.88	0.91

Table 5: Median retention and spurious signal of Gaussian transients by width L, $R = 0$, test years.

	Retention				Spurious signal			
Filter	10	25	50	100 km	10	25	50	100 km
C1	1.00	0.98	0.94	0.83	0.00	0.01	0.04	0.12
M2-self	0.75	0.24	0.03	0.00	0.43	0.31	0.17	0.11
Far stack	1.00	0.98	0.94	0.83	0.00	0.01	0.04	0.12
D1	1.00	1.00	1.00	1.00	0.01	0.04	0.11	0.25
M1	0.77	0.25	0.05	0.00	0.20	0.12	0.06	0.02
M1, 64	0.90	0.53	0.18	0.03	0.11	0.18	0.10	0.04
M1, 256	0.97	0.84	0.52	0.20	0.03	0.13	0.22	0.11
M2	0.82	0.28	0.04	0.00	0.50	0.33	0.19	0.12
M3a	0.83	0.39	0.21	0.16	0.20	0.13	0.06	0.04
M3b	0.78	0.30	0.11	0.07	0.21	0.16	0.07	0.04
M4a	0.88	0.52	0.39	0.34	0.17	0.12	0.06	0.04
M4b	0.78	0.33	0.16	0.11	0.20	0.14	0.07	0.04
M3k	0.77	0.25	0.04	0.01	0.22	0.15	0.06	0.03
M3k, 256	0.88	0.70	0.45	0.20	0.05	0.09	0.22	0.18
M5	0.95	0.83	0.72	0.59	0.08	0.09	0.09	0.16
M5-aug	1.00	0.98	0.93	0.82	0.02	0.03	0.07	0.16

Table 6: C3 settings (k regions, p components), $R = 0$, test years.

Setting	Noise removed	Kept 25 / 100 km	Spurious	Afterslip	Offsets	Ridgecrest
$k = 1$, $p = 1$	22.7%	0.99 / 0.85	1.58	1.00	0.87	0.92
$k = 1$, $p = 2$	23.1%	0.98 / 0.82	0.58	1.00	0.87	0.92
$k = 1$, $p = 3$	21.9%	0.98 / 0.81	0.57	1.00	0.87	0.92
$k = 4$, $p = 1$	19.8%	0.95 / 0.58	0.39	0.99	0.82	0.94
$k = 4$, $p = 2$	19.5%	0.94 / 0.56	0.43	0.99	0.78	0.94
$k = 4$, $p = 3$	19.0%	0.93 / 0.54	0.46	0.99	0.78	0.93
$k = 8$, $p = 1$	16.2%	0.90 / 0.35	0.24	0.99	0.68	0.84
$k = 8$, $p = 2$	19.1%	0.88 / 0.32	0.28	0.97	0.67	0.83
$k = 8$, $p = 3$	19.1%	0.86 / 0.31	0.29	0.97	0.69	0.79

Table 7: Afterslip retention per station by distance from the slipping patch (median), and the shallow slow-slip events, $R = 0$, test years.

	Afterslip, distance from patch				Shallow slow slip	
Filter	0–5	5–15	15–40	over 40 km	Kept	Spurious
C1	1.00	1.00	1.00	0.98	1.00	0.00
M2-self	0.94	0.78	0.18	−0.04	0.98	0.21
Far stack	1.00	1.00	1.00	0.98	1.00	0.00
D1	1.00	1.00	1.00	1.00	1.00	0.00
M1	0.97	0.89	0.11	−0.05	1.00	0.09
M1, 256	1.01	1.00	0.96	0.75	1.00	0.01
M2	0.96	0.85	0.20	−0.04	0.99	0.27
M3a	0.99	0.89	0.23	0.06	0.99	0.09
M3b	0.97	0.89	0.24	0.08	0.99	0.07
M4a	1.01	0.96	0.54	0.41	1.00	0.09
M4b	0.96	0.86	0.33	0.19	0.99	0.08
M3k	0.99	0.85	0.13	−0.05	0.97	0.08
M3k, 256	0.99	0.94	0.72	0.47	0.97	0.02
M5	1.00	0.98	0.90	0.77	0.99	0.03
M5-aug	1.00	1.00	0.99	0.97	1.00	0.01

Table 8: Paired bootstrap against the far stack, $R = 0$, test years: median reconstruction error and the 95% interval of the difference (filter minus far stack). Seen: 936 stations; held-out: 163; all: 1,099; 58 blocks of 30 days.

Filter	Seen	Held-out	All
Far stack	0.5632	0.5569	0.5622
M5	0.5619 [−0.0061, +0.0013]	0.5470 [−0.0174, +0.0077]	0.5601 [−0.0067, +0.0010]
M5-aug	0.5632 [−0.0027, +0.0013]	0.5543 [−0.0062, +0.0021]	0.5625 [−0.0027, +0.0012]
M3k, 256	0.5654 [−0.0089, +0.0101]	0.5522 [−0.0281, +0.0255]	0.5636 [−0.0092, +0.0092]

Table 9: Velocity repeatability: median $|\Delta v|$ (mm/yr) between 2022–2023 and 2024–2025, $R = 0$ unless stated.

Filter	East	North	Up
No filter	1.98	0.82	4.59
C1	0.83	1.53	2.41
C3 (9 settings)	0.86–1.43	0.79–1.41	1.62–2.16
M2-self	0.81	0.87	1.48
Far stack	0.83	1.53	2.42
D1	0.85	2.50	2.56
M1	1.15	1.17	2.27
M1, 256	1.03	1.31	2.18
M2	0.90	1.04	1.70
M3a / M3b	1.06 / 1.10	1.06 / 1.04	2.03 / 2.31
M4a / M4b	1.24 / 1.16	0.98 / 1.05	2.41 / 2.45
M3k / M3k, 256	0.87 / 0.87	1.01 / 1.04	1.96 / 1.59
M5 / M5-aug	0.73 / 0.85	1.48 / 1.47	2.17 / 2.17